

How Artificial Intelligence Transforms Discovery, Monetization, and Live Commerce on Fashville

A full technical case study covering problem framing, production architecture, experimentation, results, platform leverage, and governance.

Market Context: Why This Platform, Why Now

The African fashion market is not a niche. It is a \$31 billion industry growing at approximately 10 percent annually, with diaspora spending in the United States, United Kingdom, and Canada adding a further \$2.5 billion in cross-border demand that existing global marketplaces have not meaningfully served. The structural gap is on both sides. African fashion creators, from Lagos Ankara tailors to Accra streetwear labels, have no dedicated platform that understands their aesthetic vocabulary, supports African mobile payment infrastructure, or protects them through escrow on cross-border transactions. African and diaspora buyers searching for occasion-specific, culturally specific garments are underserved on every general-purpose marketplace because those platforms were not built with their vocabulary, their occasions, or their aesthetics in mind.

Fashville enters as a vertical two-sided marketplace built specifically for this gap. The creator supply side is large, fragmented, and currently monetizing through Instagram DMs, WhatsApp, and informal logistics with no buyer protection and no discovery infrastructure. The buyer demand side is high-intent, willing to pay premium prices for culturally resonant garments, and deeply frustrated by the search experience on platforms that return irrelevant results for queries like Owambe ready co-ord or modern Kente blazer. A platform that closes the vocabulary gap and builds trust infrastructure on both sides is not competing with Etsy or ASOS. It is opening a market that has not been properly served before. The AI platform described in this case study is not a feature differentiator. It is the mechanism that makes the two-sided market work at scale.

Executive Summary

As Fashville scales, discovery systems optimized for engagement signals produce ranking behavior that does not consistently align with buyer intent or creator monetization goals. On a platform where African cultural aesthetic vocabulary is central to the product, the gap between

what buyers mean and what traditional keyword systems understand is structurally large and structurally growing as the creator base expands.

This case study describes the design of a semantic relevance and AI intelligence platform that introduces LLM-aligned scoring anchored to human ground truth, distills semantic reasoning into production-grade models, integrates multimodal signals across text, image, video, and live streams, reduces cold start friction for emerging creators, and becomes a reusable AI infrastructure layer across all ranking and recommendation surfaces.

Central to everything is a proprietary African fashion ontology built before any model was selected. It is the asset that makes the platform's AI defensible. General-purpose models trained on standard corpora do not understand what Owambe ready means in a fashion context. The ontology does. Every downstream model benefits from it permanently.

The Platform Problem

The system was not broken. It was optimizing for the wrong thing. Four failure modes were mapped before a single requirement was written.

Gap 01: Engagement dominated ranking

Clicks and saves outweighed purchase intent signals. Popular items were not necessarily relevant. Cart abandonment rose while mismatch returns climbed.

Gap 02: Cultural vocabulary invisible to search

Queries for Owambe ready co-ord or modern Ankara blazer returned nothing useful. Lowest conversion occurred on the platform's highest intent queries. The vocabulary gap was not a minor inconvenience. It was a structural revenue problem.

Gap 03: Live commerce lacked real-time intelligence

Creators streamed without optimal timing guidance, audience targeting, or dynamic pricing insights. Live conversion rates plateaued.

Gap 04: Visual-text mismatch drove returns

Generically captioned products did not match what buyers saw in videos or live streams. Surprise at checkout eroded buyer trust and margins.

All four failure modes pointed to the same root cause: no shared semantic understanding. The ranking and recommendation systems understood behavior but not intent. They knew what people clicked. They had no idea what people meant. African aesthetic vocabulary like Owambe, Adire casual, or Kente formal was absent from standard training corpora.

Product Strategy

The first instinct was to patch each gap separately. But the right question was: if the same underlying gap appears across search, feed, live streams, and recommendations, why build four separate solutions? Build the understanding once. Serve it everywhere.

Strategic goal: build a semantic relevance and AI intelligence platform that functions as foundational infrastructure, a shared scoring and recommendation service consumed across all surfaces rather than a one-off feature improvement.

Five principles shaped every downstream decision.

01. Human-anchored relevance. AI models without human ground truth diverge from real buyer intent. Build the dataset first. Build the ontology alongside it.

02. AI offline, not in live serving paths. LLM inference at query time or during live streams violates latency and cost requirements. This is non-negotiable. Offline restriction also means annotation quality can be improved independently of serving infrastructure.

03. Distill into production models. Transfer AI quality into fast, cheap serving models so quality and speed coexist.

04. Multimodal from day one. Fashion relevance lives at the visual-textual intersection. Text alone is insufficient. Live video requires frame-level analysis. The annotation layer must process images directly, not rely on seller-provided captions.

05. Platform leverage. Build once. API-expose. All surfaces benefit. Focus on infrastructure, not isolated features.

The African Fashion Ontology: The Strategic Foundation

Before any model was selected, a foundational decision was made that shapes the entire architecture.

The African fashion ontology is a proprietary, versioned knowledge base covering aesthetic categories, occasion types, garment attributes, silhouette vocabulary, fabric signal mappings, and regional style signals specific to African fashion contexts. It was built through structured LLM-assisted taxonomy generation combined with cultural context annotators who understand the difference between traditional and fusion Ankara, Owambe-appropriate styling, Kente formal versus Kente casual, and the formality gradients within African streetwear.

This ontology is not a dataset. It is a platform asset. No competitor can replicate it without the same annotator investment and cultural grounding. General-purpose training corpora contain almost none of this vocabulary. Standard crowdsourced annotation would not have produced it. The ontology is what makes Fashville's AI defensible at a structural level, and every downstream model in the architecture is permanently better because it exists.

The ontology is versioned. When new aesthetic categories emerge, such as Afrofuturism now requiring dedicated labels where it did not two years ago, the ontology is updated first. Model retraining follows from the ontology, not the other way around.

The competitive moat argument is worth stating explicitly. Replicating this ontology requires cultural context annotators who understand African fashion vocabulary at the level of occasion, region, and aesthetic tradition. It requires annotator recruitment, training, and quality management infrastructure. It requires time. A well-resourced competitor starting today would need a minimum of 12 to 18 months to reach comparable ontology depth, and that estimate assumes they prioritize it from day one rather than treating it as a supporting asset. In practice, most competitors would attempt to close the gap through general-purpose fine-tuning on publicly available African fashion content, which would produce partial vocabulary coverage with no mechanism for structured improvement. The ontology compounds. Every labeling cycle, every new creator uploading inventory with culturally specific tagging, and every buyer interaction with culturally specific queries adds signal that refines it further. A competitor cannot buy their way into that compounding effect after the fact.

System Architecture: Three Layers, Three Jobs

The architecture has one governing idea: separate the quality problem from the latency problem.

Large language models understand what Owambe ready means in a fashion context. They know it implies celebratory occasion, African formal aesthetic, statement color, and specific silhouettes. Keyword matching cannot produce this. But calling a large language model at query time on a live marketplace is also not viable. Latency and cost make the platform unusable at scale. Knowledge distillation solves this. The large model teaches a smaller model how to reason about fashion intent. The smaller model carries that reasoning into production at sub-20ms latency.

Three layers, cleanly decoupled: reasoning offline, distillation at batch, serving in real time.

Full system architecture:

Human-labeled golden datasets across query-to-product, video-to-product, occasion-to-garment, and caption-to-attribute dimensions feed the foundation. The African fashion ontology is injected via RAG into every LLM annotation call. Claude 3 Opus and

GPT-4V operate as the offline annotation layer with version-controlled prompt templates, few-shot examples, and chain-of-thought formatting. An annotation quality gate validates every batch against the teacher model before any output enters training. The teacher model, fine-tuned from DeBERTa-v3-large, handles batch scoring and quality oracle functions. The student production model handles real-time serving at sub-20ms P95 latency. The student model is consumed by the search ranker, feed ranker, live stream optimizer, AI shopping assistant, pricing engine, and creator caption generation tool.

Data and Modeling Stack

Layer 1: Human Ground Truth

Golden datasets across four dimensions: query-to-product relevance, live video-to-product consistency, caption-to-attribute alignment, and occasion-to-garment fit. Structured relevance labels use a graded scale. Investment in labeling infrastructure came first, before any model selection. This dataset anchors all downstream quality.

Labeling required annotators with cultural context who understand Owambe-appropriate styling, traditional versus fusion Ankara, and the formality spectrum within African streetwear. Generic crowdsourced annotation would have failed here because the vocabulary gap is cultural, not linguistic.

Layer 2: LLM Annotation Layer with RAG

Claude 3 Opus handles complex attribute inference and cultural vocabulary resolution. GPT-4V handles image-to-attribute annotation tasks. Both models are restricted to offline operation. They never touch production traffic.

Standard LLM prompting against these models would partially close the vocabulary gap. It would not close it fully. The reason is that general-purpose training corpora contain limited African fashion vocabulary. The fix is Retrieval-Augmented Generation.

Before any annotation call, the pipeline queries the African fashion ontology and injects the relevant vocabulary context directly into the prompt. A query for Owambe ready co-ord triggers retrieval of occasion classification rules, appropriate silhouette definitions, color signal guidelines, and fabric attribute mappings from the ontology. Claude 3 Opus annotates with that context in its window, not from generic training alone. This is what closes the vocabulary gap structurally.

For image and video annotation tasks, GPT-4V receives the product image or video frame alongside the structured prompt. The model infers silhouette category, print density, color distribution, and cultural style signals directly from visual input rather than relying on

seller-provided captions. This closes the gap between what buyers see and what the catalog describes without requiring additional creator effort.

Prompt engineering is version-controlled alongside model versions. Each annotation task type, including attribute inference, occasion classification, visual-text alignment scoring, and caption quality assessment, has a dedicated prompt template with few-shot examples drawn from the human-labeled golden dataset. Chain-of-thought formatting is used for complex inferences so annotation outputs are inspectable and auditable. When the underlying model updates, prompt regression tests run automatically against a held-out sample of golden labels before any annotation batch proceeds.

Annotation Quality Gate

LLM inference generates errors. The architecture treats this as a known variable, not an edge case. Every annotation batch passes through a quality gate before it enters the training dataset. The gate runs the teacher model as a quality oracle against a sampled subset of LLM annotations. Batches that fall below a precision threshold on the graded relevance scale are flagged for human review rather than passed downstream. Bad annotations fed into distillation training corrupt the student model. The quality gate is what keeps annotation quality and serving quality coupled across every model update cycle.

Layer 3: Teacher Model

A mid-size transformer fine-tuned from DeBERTa-v3-large as the base, trained via supervised fine-tuning to match LLM annotation outputs across all four labeling dimensions.

The choice to fine-tune a base model rather than prompt-engineer a frontier model for the teacher role was deliberate. Fine-tuning produces a model that is faster, cheaper to run at batch scale, and fully owned. The tradeoff is that updating the teacher requires a retraining run when LLM annotation quality improves significantly. That cost is acceptable because teacher updates happen on a quarterly cycle aligned with the ontology versioning schedule.

The teacher runs batch scoring weekly on new catalog items and serves as the quality oracle during ranker training.

Cost architecture for the annotation layer: At approximately 350,000 items annotated per week with an average of 600 tokens per item, weekly LLM annotation cost runs to approximately \$180 at current API pricing. As the catalog scales to one million items, that cost reaches approximately \$500 per week. Serving costs are entirely decoupled from this because the student model handles all real-time queries. This layering keeps serving costs independent of annotation quality decisions.

Layer 4: Student Production Model

A compact transformer distilled from the teacher. Sub-20ms P95 latency. High QPS. Memory-efficient via quantization. Outputs relevance scores, style confidence vectors, visual-text alignment scores, and live commerce engagement predictions.

The serving SLA is a strict product requirement. A small precision loss relative to the teacher was accepted in exchange for a 10x latency improvement. The quality loss occurs in the tail of the distribution. High-relevance items maintain strong scores across all query types including culturally specific vocabulary.

Multimodal Feature Pipeline

Fashion relevance cannot be resolved from text alone. Structured minimalist blazer implies a specific silhouette, texture, and color density. These signals live in images, videos, and live stream frames.

Text encoders process queries and captions. Image encoders extract silhouette, texture, and color distribution. Video and live encoders handle a challenge unique to live commerce: streams run for one to three hours. The transcript pipeline uses a sliding window chunking strategy with 512-token segments and 128-token overlap. Frame sampling occurs every 15 seconds during encoding rather than continuously. Real-time chat sentiment is tracked through a separate lightweight classifier operating on a 60-second rolling window. This produces a temporally grounded engagement signal without creating a context window bottleneck.

An attention fusion block learns cross-modal alignment weights dynamically. For bold print co-ord queries, the image signal dominates. For breathable fabric casual dress queries, text and attribute signals dominate. The attention weights learn this from the labeled dataset specifically for African aesthetic vocabulary.

Creator caption and hashtag generation consumes this same semantic infrastructure rather than being built as a separate AI feature. When a creator uploads a product, the visual-text alignment score from the student model drives two outputs: a quality flag if the existing caption falls below the alignment floor, and a suggested caption generated by calling Claude 3 Opus with the product image, inferred attributes from GPT-4V, and a generation prompt template drawing vocabulary from the ontology. The same infrastructure that powers search relevance powers the creator-facing tool. No separate AI development was required.

Ranking and AI Integration

Four integration points restructure how the system learns intent over time.

Point 01, Candidate filter: Hard pruning before heavy ranking. Removes obvious mismatches to control compute cost.

Point 02, Feature engineering: Relevance serves as the primary ranking feature, crossed with price, style embeddings, and live engagement signals.

Point 03, Loss weighting: High semantic relevance pairs contribute more to training loss. The ranker learns intent structurally rather than additively.

Point 04, Cold start and live commerce boost: Strong semantic match unlocks visibility for new creators and first-time streamers. Marketplace fairness does not happen by accident. Semantic matching gives quality a path to visibility that does not require historical engagement. Supply side health drives long-term GMV. A platform that suppresses new creators until they overcome discovery barriers will never close the cold start gap.

Measuring What Actually Matters

New platform metric: Semantic Purchase Alignment (SPA)

Definition: percentage of top 5 results rated highly relevant by the teacher model for a given query cohort. Updated daily. Excludes queries with fewer than 10 impressions in the rolling window.

SPA is a leading indicator of intent alignment upstream of conversion. Conversion tells us if buyers are transacting. SPA tells us if what was shown to them was actually correct. SPA is tracked alongside live stream-to-purchase conversion, creator subscription upgrade rate, return rate, creator exposure diversity index, and latency P95.

Experimentation Strategy

Four phases with hard guardrails at each gate. No phase advanced without clearing all three guardrails: latency P95 below threshold, no GMV degradation, and diversity index above floor.

Phase 1, Offline evaluation: Teacher model scored against human labels. Precision and recall targets were set. Query buckets with the largest semantic gap were identified, specifically occasion-based, cultural vocabulary, and long-tail queries. Early bias signals surfaced where the model under-scored products tagged with certain regional attributes. Fixed during the next labeling refresh using the ontology as the correction reference.

Phase 2, Limited query and live stream A/B test: Student model deployed on a curated subset targeting occasion-based and cultural vocabulary queries. Live stream optimization

tested on a subset of creators. Measured conversion, latency, return rate, and live engagement. GMV guardrail held. Latency P95 stabilized at 18ms. Diversity index remained stable.

Phase 3, Full integration: Relevance scores and AI recommendations deployed across all search, feed, and live surfaces. Active diversity index guardrails introduced to prevent aesthetic concentration around dominant style clusters.

Phase 4, Platform API exposure: Semantic scoring and AI recommendation service exposed internally. The AI shopping assistant, pricing insights, trend forecasting, and live stream tips began consuming the same underlying service. The infrastructure flywheel activated.

Results: 90-Day Window Post Launch

Percentages without business context are incomplete. The numbers below are translated against a baseline monthly GMV of \$1.2 million at the point of deployment, with an average order value of \$85 across the platform.

Search-to-purchase conversion rate: plus 18 percent

At baseline, search-driven sessions converted at approximately 3.1 percent. An 18 percent uplift brings that to 3.7 percent. Across the monthly search session volume at deployment, this translates to roughly \$216,000 in incremental monthly GMV attributable to improved relevance alone. Annualized, that is approximately \$2.6 million in GMV recovered from a discovery gap that existed before any AI investment.

Mismatch-driven return rate: minus 22 percent

Returns on a fashion marketplace carry a full reversal cost including logistics, restocking, and escrow release processing. At a conservative estimate of \$12 per return event, a 22 percent reduction in mismatch returns across the platform's monthly return volume represents approximately \$38,000 in monthly cost savings. Over 12 months, that exceeds \$450,000 in cost recovery. The buyer trust implication compounds beyond direct cost. Repeat purchase rate increases as return-driven churn decreases, and buyer LTV improves without any additional acquisition spend.

New creator discovery: plus 14 percent uplift

New creator visibility directly influences supply-side health. A creator who earns their first meaningful sales within 30 days of joining is significantly more likely to remain active, upload inventory consistently, and upgrade their subscription tier. The 14 percent uplift in new creator discovery, driven by semantic match replacing pure engagement-signal ranking, expanded the active creator base and initiated the upgrade conversion funnel at an earlier stage of the creator lifecycle.

Live commerce conversion and creator subscription upgrades

Live stream conversion improvements, combined with AI-powered broadcast timing and pricing insights available at the Pro and Premium tiers, produced a measurable increase in subscription upgrade rates. The specific revenue implications of subscription upgrades are addressed in the creator economics section below.

Signal strength was highest for occasion-based and culturally specific queries. Buyers searching for bridesmaid looks for outdoor Yoruba introduction ceremonies expressed layered, multi-dimensional intent. The AI platform resolved it. Keyword matching could not. High-intent, culturally specific queries carry average order values 30 to 40 percent above the platform average because occasion dressing is high-commitment purchasing. Conversion uplift concentrated in this query cohort compounded into the GMV impact described above.

Creator Economics: How AI Investment Becomes Subscription Revenue

Fashville's primary revenue engine is not transaction commission. It is subscription tier upgrades, where commission rates drop from 8 percent on Free to 5 percent on Pro and 3 percent on Premium while monthly fees of \$9.99 and \$29.99 respectively shift the revenue mix toward predictable recurring income. Understanding this changes how the AI investment is evaluated commercially.

Consider a creator generating \$5,000 in monthly GMV. On the Free tier, Fashville earns \$400 in commission. On the Pro tier at \$9.99 per month, Fashville earns \$250 in commission plus \$9.99 in subscription revenue, totaling \$259.99. That is a 35 percent reduction in revenue per creator at this GMV level if the subscription fee is the only addition. The business only wins when creators on Pro and Premium tiers generate significantly higher GMV than they would on Free, which is exactly what the AI features are designed to produce. AI pricing insights, trend forecasting, and featured live placement are not cosmetic upgrades. They are GMV growth tools. A creator who upgrades to Premium and uses AI trend forecasting to stock ahead of regional demand cycles generates more GMV, which benefits both the creator and the platform even at the lower 3 percent commission rate. At \$15,000 monthly GMV on Premium, Fashville earns \$450 in commission plus \$29.99 in subscription revenue. That is a better outcome than \$400 from the same creator capped at \$5,000 GMV on Free.

The cold start strategy reinforces this economics chain at the beginning of the creator lifecycle. A new creator who gets early visibility through semantic matching earns their first sales faster. Faster first sales shorten the time to Pro upgrade consideration. The 14 percent uplift in new creator discovery is therefore not just a fairness metric. It is the top of the subscription upgrade funnel moving faster. Every new creator who reaches \$2,000 monthly GMV within their first 60 days is a viable Pro upgrade candidate. At scale, compressing that timeline by two to three

weeks across hundreds of new creators per month produces a measurable shift in monthly recurring subscription revenue. The AI platform does not just improve discovery. It accelerates the entire creator monetization journey from first sale to subscription commitment.

Platform Leverage: From Feature to Infrastructure

The single most important outcome was not in the search metrics. It was what happened when relevance and AI scores were exposed as an internal API.

The AI Shopping Assistant consumed semantic scoring for style matching and outfit pairing.

The Live Streaming team used engagement predictions for broadcast timing and featured product placement.

The Creator Monetization team integrated AI pricing insights and trend forecasting into the Pro and Premium subscription tiers.

The Catalog team used visual-text alignment scores to surface poorly tagged listings to creators at upload time, with the caption generation tool closing the loop.

Each of these would have required separate AI development if built as isolated features. As infrastructure, they scaled at near-zero marginal cost. The African fashion ontology sits underneath all of them. Updating the ontology improves every surface simultaneously.

The Hardest Decisions

Decision 01, LLM placement: Chose Claude 3 Opus and GPT-4V for offline annotation only. Rejected LLM inference in serving paths. Latency and cost would have made the platform untenable at production scale. Offline restriction enables continuous annotation quality improvements without disrupting serving infrastructure. It also keeps costs bounded and forecastable.

Decision 02, Model architecture: Chose fine-tuning DeBERTa-v3-large as the teacher rather than prompt-engineering a frontier model. Fine-tuning produces a fully owned model with faster batch scoring and predictable costs. The retraining overhead on quarterly cycles is acceptable. Chose teacher-to-student distillation over a single large model end to end. Distillation decouples quality iteration from serving cost. Improve the teacher, re-distill, deploy the new student with no serving infrastructure changes required.

Decision 03, RAG for vocabulary closure: Chose RAG-injected ontology context for every annotation call rather than relying on general-purpose model training. Relying on training data

alone would have produced partial vocabulary coverage with no mechanism for improvement. The ontology gives the annotation layer a stable, versioned, culturally grounded vocabulary source that improves as annotators expand it.

Decision 04, Creator monetization and cold start strategy: Chose semantic match as a quality proxy for new creators. Rejected pure historical signal reliance. Semantic matching aligns business incentives with marketplace fairness.

Decision 05, Bias monitoring scope: Chose explicit audits across body size, region, and aesthetic category. Rejected aggregate metrics only. Fashion platforms have documented representation failures. Aggregate metrics mask systemic suppression of plus-size, darker-skinned, or regionally specific products. Disaggregated metrics are a product quality requirement, not a compliance checkbox.

Model Governance

An AI relevance platform that is not actively maintained degrades. Embedding distributions shift as the catalog grows. Fashion trends move. New African aesthetic categories emerge. Afrofuturism requires dedicated labels today where it did not two years ago. The ontology must be updated first. Models follow.

Governance is a roadmap item, not a postmortem action.

Weekly embedding distribution drift detection, scheduled as a shipping milestone.

Quarterly ontology versioning with cultural context annotators, followed by annotation refresh runs and teacher retraining.

Bias audits across body size, region, and aesthetic category on a quarterly cycle.

Prompt regression tests triggered on every underlying model update before any annotation batch proceeds.

Student model retraining triggered on teacher updates.

Feature store integration for reproducibility.

Embedding cache invalidation on model updates.

Model quantization for serving cost control.

Batch annotation job scheduling and monitoring.

New aesthetic category detection via clustering fed back into ontology proposals for annotator review.

Visual-text alignment floor for catalog quality gating enforced at upload time.

Cost architecture remains layered by purpose. Claude 3 Opus and GPT-4V handle offline annotation only with bounded, predictable costs scaling from \$180 to approximately \$500 per week as the catalog grows to one million items. The teacher model runs batch scoring weekly on new catalog items. The student model handles real-time serving with quantized memory efficiency. This layering keeps serving costs decoupled from annotation quality decisions permanently.

ROI framing on the full AI investment: Total annual annotation and infrastructure cost at current scale runs to approximately \$140,000, covering LLM annotation, teacher batch scoring, student serving infrastructure, and labeling operations. Against the \$2.6 million in annualized incremental GMV from conversion improvement alone, and \$450,000 in annualized cost recovery from return rate reduction, the AI platform generates approximately \$22 in platform value per dollar of AI investment in year one. That ratio improves as the catalog scales because annotation costs grow sub-linearly relative to GMV growth, and the ontology appreciates in value without proportional additional investment.

What Doing All Three Together Actually Teaches You

Most PM assessments test one dimension at a time. This challenge tested all three simultaneously and forced a coherence that isolated assessments cannot achieve.

When you build the prototype, write the PRDs, and design the AI architecture for the same platform at the same time, nothing can be inconsistent. The creator equity and monetization thinking in the AI case study must surface in the dashboard monitoring layer. The AI shopping and search requirements in the PRD must match the architecture in the case study. The dashboard views must reflect the metrics defined elsewhere. That constraint is where real product thinking happens. Not in any individual deliverable. In the connections between them.

01. Platform choice is a product decision. Choosing African fashion creators identified a real two-sided market gap, a real AI problem regarding cultural vocabulary underrepresentation, and a real infrastructure opportunity. The ontology is the proof.

02. The dashboard is a product, not a report. It has users, jobs to be done, a north star, and acceptance criteria. Treating it as an afterthought produces unused tools.

03. PRDs fail when they start with features. The requirement is downstream of the problem. Lead with who experiences what problem at what cost. Let features follow from that.

04. Architectural decisions in an AI case study are PM decisions. Where the LLM sits, what it annotates, what it never touches, whether RAG is needed, how prompts are versioned, what the annotation quality gate accepts, these have direct cost, latency, and quality implications. Own them.

05. Governance is a roadmap item. If weekly drift detection, quarterly ontology updates, and prompt regression testing are not on the roadmap, they will not happen. The model will degrade, the vocabulary coverage will drift, and metrics will slip.

06. Platform thinking is the unlock. The AI relevance service started as a search and live commerce improvement. It became infrastructure for multiple teams. The African fashion ontology started as a labeling aid. It became the platform's deepest competitive moat. That is the difference between a PM who ships features and a PM who builds foundations.

I came into this challenge knowing how to build products. I left knowing how to architect platforms and understanding the difference between the two. Not just knowing what an AI model is. Knowing where intelligence belongs in a production system. Knowing how to separate the quality problem from the latency problem. Knowing how to make AI infrastructure, including the proprietary knowledge assets underneath it, reusable across surfaces rather than rebuilt for each one. That is what a Platform PM at an AI-first company actually requires. And this is how I built the case for it.